

Two Measures on One Spring

How Much, Not Where: Invariant and Variant Averages Under Reorientation in a Gravitational Field

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Abstract

A measure is a rule for saying how much of an object lies in any chosen part of it. A coiled spring carries two such rules that a student can hold in their hands: count the coils in a portion, or measure the centimetres that portion occupies. Lying flat the two rules agree up to a constant and nothing built on either can tell them apart. Hung vertically, the spring stretches non-uniformly under its own weight, the two rules come apart, and the consequences are exact. The mean and median position separate under each rule into three distinct points rather than four, because the length-mean and length-median coincide. The average pitch computed by counting differs from the average computed by ruler, and the discrepancy is precisely the gap between the harmonic and arithmetic means of the pitch under the length rule, with the count-average equal to the harmonic length-average identically. The local pitch, the visible distance between adjacent coils, is the conversion rate between the two rules, and is the object that measure theory later calls a Radon–Nikodym derivative. The development proceeds from a finite, countable stage before any integration, separates a change in the weighting rule from a change in the quantity being weighted, and takes the convergent field, where weight detaches from mass as a third rule, as an independent extension.

1 The spring, flat and hanging

A long soft spring lying flat on a bench, carrying no axial tension, has evenly spaced coils. Its midpoint found by folding, its balance point found by suspending it from a thread, and its middle coil found by counting all fall in the same place. Nothing distinguishes them and nothing yet needs a name.

Hang the same spring from one end and the coils space out unevenly, widely near the support and tightly near the free end. The three points that coincided now lie at three different heights, and two reasonable procedures for computing the average gap between neighbouring coils return two different numbers. Both facts follow from a single circumstance: the spring carries two distinct rules for saying how much of it lies in a given portion, and stretching has pulled those rules apart.

How pronounced this is depends on a single dimensionless number, the spring's own weight W_s set against the force kL_0 that would double the natural length L_0 of a spring of constant k :

$$\lambda \equiv \frac{W_s}{kL_0}.$$

A stiff, light spring has λ near zero and hangs almost as evenly as it lies; a slinky, whose own weight extends it many times over, has λ in the tens. The hanging length is $L = L_0(1 + \frac{1}{2}\lambda)$

and the pitch at the support exceeds the pitch at the free end by the factor $1 + \lambda$, both derived in Section 4. Figure 1 shows a spring with $\lambda = 8$, exaggerated well beyond anything a laboratory spring would show.

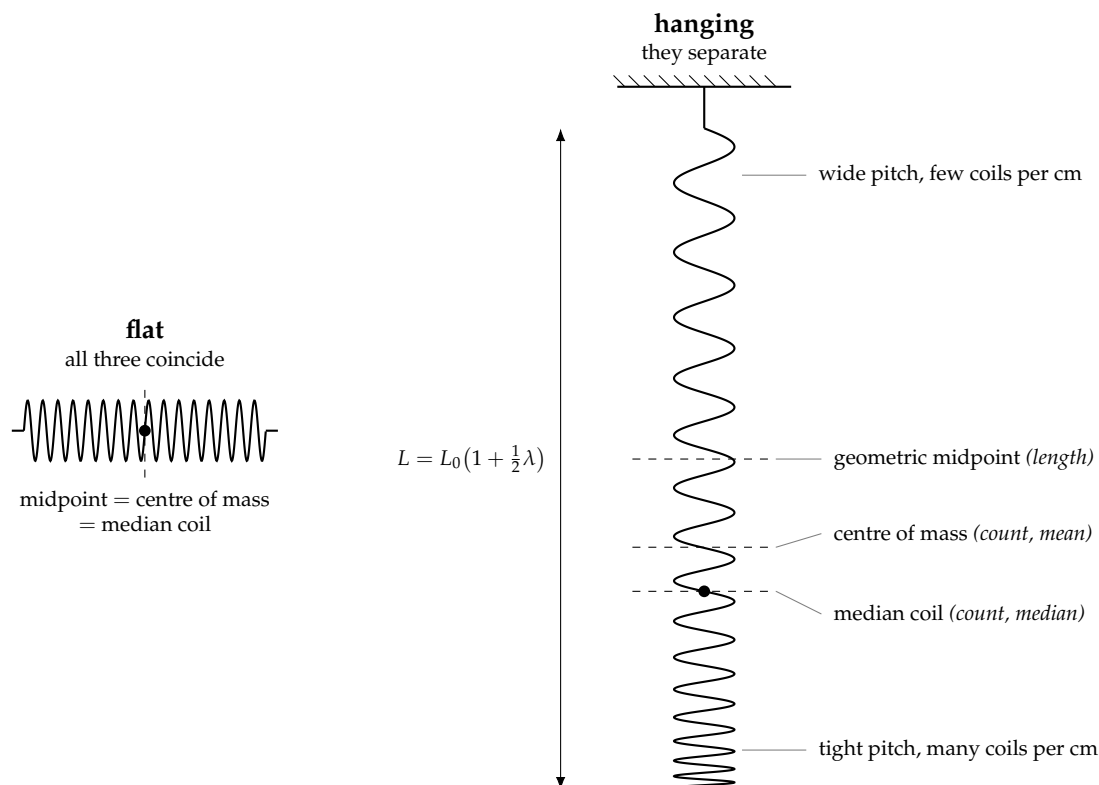


Figure 1: A very floppy spring, $\lambda = 8$, drawn to scale from its own stretch profile. Flat, the coils are evenly spaced and the three marks fall together. Hanging, the coils near the support are pulled far apart while those near the free end stay tight, and the three marks separate in a fixed order. A real laboratory spring has a much smaller λ and shows a much smaller version of the same separation.

Dividing the hanging spring into blocks of equal coil count and measuring the length each block occupies makes the divergence concrete. The blocks are equal under one rule and unequal under the other, and every result below is a consequence of that inequality. The classroom sequence that develops this material in the order a learner needs it is given in the appendix and in the accompanying student investigation [3].

2 What a measure is

A measure is a rule for saying how much of an object lies in any chosen part of it. The spring offers two.

For any chosen portion P of the spring, let

$\mu_{\text{count}}(P)$ = the number of coils lying in P , $\mu_{\text{len}}(P)$ = the length, in centimetres, that P occupies.

Both deserve the name because both are additive: if P and Q are portions that do not overlap, then

$$\mu(P \cup Q) = \mu(P) + \mu(Q),$$

which is the defining property. Blocks of the spring laid end to end have coil counts that sum to the total coil number and lengths that sum to the total length, so both rules qualify.

A measure is not a coordinate. A coordinate says *where* you are: the distance x from the support, or the coil number n counted from it whereas a measure says *how much* is in a region.

The two rules disagree in the following concrete sense. Choose a coil at random, each coil equally likely; that is sampling under μ_{count} . Choose a centimetre at random along the ruler, each centimetre equally likely, and look at whichever coil is there; that is sampling under μ_{len} . On the flat spring these are the same procedure in different words. On the hanging spring they are not, because the stretched coils near the support occupy more ruler.

3 Conversion rate

Number the coils $n = 0, 1, \dots, N$ from the support and let $x(n)$ be the distance of coil n from the support. The *pitch*

$$p(n) = x(n+1) - x(n) \approx \frac{dx}{dn}$$

is the visible distance between adjacent coils at position n , in centimetres per coil. It is the rate at which one measure converts into the other: a portion containing Δn coils near position n occupies about $p(n) \Delta n$ centimetres, so

$$\mu_{\text{len}} \approx p \cdot \mu_{\text{count}} \quad \text{locally.}$$

Where the coils are far apart p is large and a coil buys a lot of ruler; where they are bunched it is small. Its reciprocal $1/p$ is coils per centimetre, and multiplying by the mass per coil m/N gives the linear mass density $m/(Np)$.

This quantity, the local conversion rate between two measures, is what measure theory calls the Radon–Nikodym derivative of μ_{len} with respect to μ_{count} [4]. The name attaches to a quantity that can be read off a hanging spring with a ruler.

4 Pitch profile

Take the spring to have natural length L_0 , N coils, total weight W_s , and spring constant k . Hung from the top, coil n supports the weight of everything below it, so the tension there is $T(n) = W_s(1 - n/N)$. Each individual coil is one of N identical springs in series, so has stiffness kN , and stretches by $T/(kN)$. The pitch is therefore

$$p(n) = \frac{L_0}{N} + \frac{W_s}{kN} \left(1 - \frac{n}{N}\right) = \frac{L_0}{N} (1 + \lambda(1 - u)), \quad u = \frac{n}{N}, \quad \lambda \equiv \frac{W_s}{kL_0},$$

falling linearly from $(L_0/N)(1 + \lambda)$ at the support to L_0/N at the free end, which is the profile drawn in Figure 1, and which confirms the factor $1 + \lambda$ between the two ends quoted in Section 1. Summing the pitches,

$$x(u) = L_0 \left(u + \lambda \left(u - \frac{1}{2} u^2 \right) \right), \quad L = x(1) = L_0 \left(1 + \frac{1}{2} \lambda \right),$$

the familiar result that self-weight extends a spring by half what an equal point load would.

Ask where the middle of the hanging spring is and there are two independent choices to make: which measure supplies weights, and which statistic is taken.

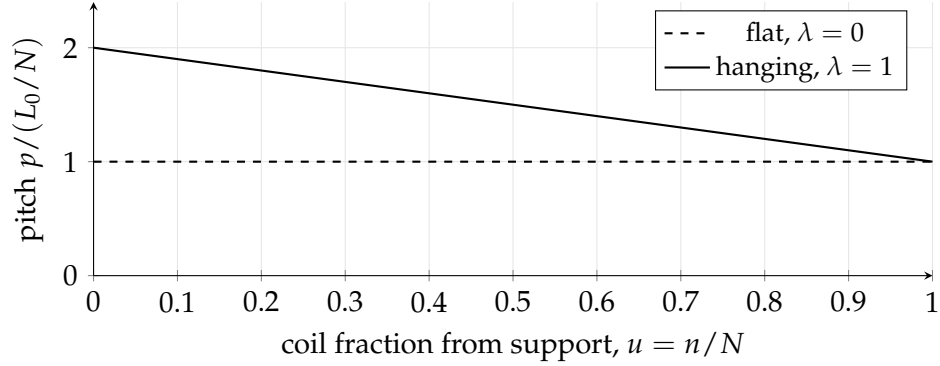


Figure 2: Conversion rate between the two measures. Flat, it is constant and the measures are proportional. Hanging, it falls linearly: coils near support each occupy more ruler than coils near the free end.

	Length measure	Count (mass) measure
Mean position	geometric midpoint, $L/2$	centre of mass
Median position	geometric midpoint, $L/2$	median coil

Table 1: The two choices behind “the middle”. The length-mean and length-median coincide because the length measure is uniform in x by construction, which is why four cells give three distinct points.

Three separated points invite the conclusion that the measure accounts for all the separation, when one of the three differences is a change of statistic instead. Evaluating,

$$x_{\text{mid}} = \frac{L}{2} = L_0 \left(\frac{1}{2} + \frac{\lambda}{4} \right), \quad x_{\text{cm}} = \int_0^1 x \, du = L_0 \left(\frac{1}{2} + \frac{\lambda}{3} \right), \quad x_{\text{med}} = x\left(\frac{1}{2}\right) = L_0 \left(\frac{1}{2} + \frac{3\lambda}{8} \right).$$

All three sit at $L_0/2$ when $\lambda = 0$ and separate linearly in λ , in that order downward. For a slinky with $L_0 = 5$ cm and $\lambda = 20$, the hanging length is 55 cm, the midpoint is at 27.5 cm, the centre of mass at 35.8 cm, and the taped middle coil at 40 cm: differences of centimetres, found with a ruler, after the order has been predicted.

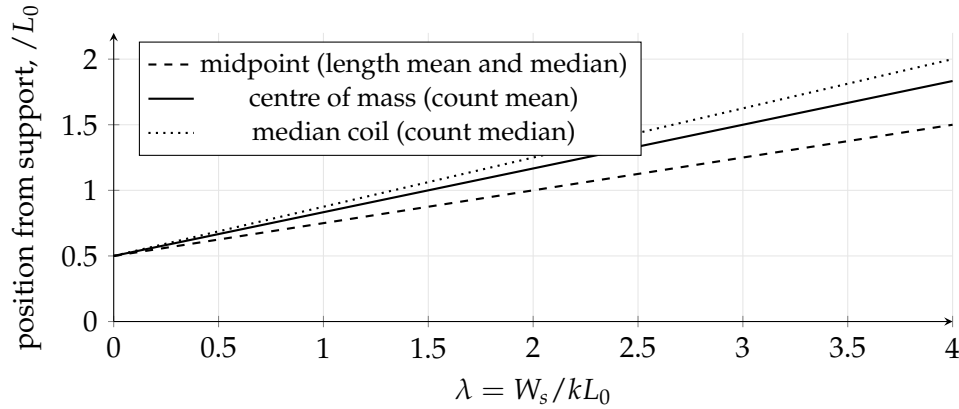


Figure 3: The three centres against the dial. They coincide only when the spring is unstretched.

5 Two averages of one pitch

Now consider the average pitch, which is where the identity lives. One student counts N coils, measures the total length, and divides. That is the arithmetic mean of the pitch under the count measure:

$$\text{AM}_{\text{count}}[p] = \frac{1}{N} \sum_n p(n) = \frac{L}{N}.$$

A second student lays a ruler along the spring, reads the local gap at every centimetre mark, and averages the readings. That is the arithmetic mean of the pitch under the length measure:

$$\text{AM}_{\text{len}}[p] = \frac{\sum_n p(n) \cdot p(n)}{\sum_n p(n)} = \frac{L_0}{N} \cdot \frac{1 + \lambda + \frac{1}{3}\lambda^2}{1 + \frac{1}{2}\lambda}.$$

These differ. By Cauchy–Schwarz the second exceeds the first whenever the pitch varies at all, with equality only for the flat spring. The reason is that the ruler samples the stretched region more often than its share of the coil population, because the stretched region is longer; this is length-biased sampling, the same effect that makes the average class size experienced by a student exceed the average class size across classes, or the average wait for a bus exceed half the average gap [5].

The count-average has a second name under the length measure, and this is the exact result rather than an analogy. The harmonic mean of the pitch under the length measure is

$$\text{HM}_{\text{len}}[p] = \frac{\sum_n p(n)}{\sum_n p(n) \cdot \frac{1}{p(n)}} = \frac{L}{N} = \text{AM}_{\text{count}}[p],$$

identically, for any pitch profile whatever, because weighting by length is weighting by p and the harmonic mean is the arithmetic mean under reciprocal weighting. The geometric mean sits between them as it must,

$$\text{HM}_{\text{len}} \leq \text{GM}_{\text{len}} \leq \text{AM}_{\text{len}},$$

the Pythagorean chain [6], collapsing to equality exactly when the spring lies flat. At $\lambda = 1$ the three take the values 1.500, 1.528 and 1.556 in units of L_0/N , and the ratio $\text{AM}_{\text{len}}/\text{HM}_{\text{len}}$ rises from 1 toward a limit of $4/3$ as $\lambda \rightarrow \infty$.

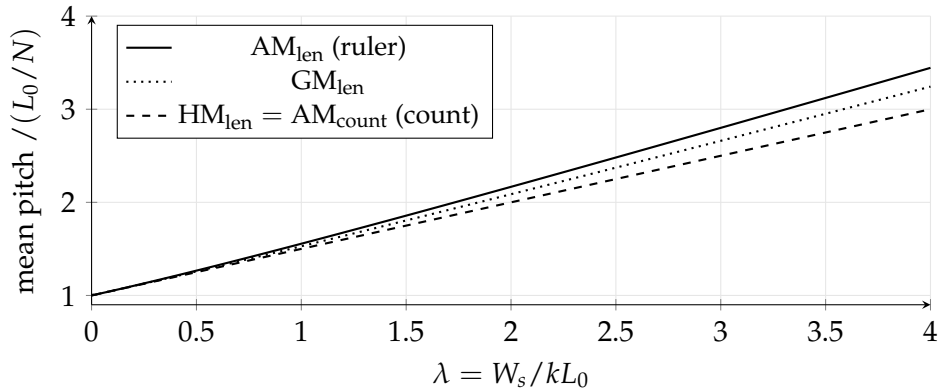


Figure 4: The three Pythagorean means of the pitch under the length measure, against the dial.

Why the ruler answers a different question

The ruler average is the correct answer to the question of what the pitch is at a typical point along the spring's length: it answers the question of the average pitch per coil wrongly only because that is a different question. An average remains incomplete until the measure supplying its weights has been named, and two people using the same word in good faith can be computing different quantities. The arithmetic mean is the familiar average and the harmonic mean the examination curiosity, yet here it is the harmonic mean that answers the question about coils.

6 Change under reorientation

The word rotation needs qualifying, because a rigid rotation changes nothing at all. Rotating a spring bodily carries the length measure with it, and every distance between coils is preserved. What happens when the spring is turned from horizontal to vertical is a two-step process: the spring is reoriented with respect to the gravitational field, and it then deforms into a new equilibrium under the tension that reorientation produces. Every effect described here belongs to the second step. It is the deformation, not the rotation, that separates the measures.

A second distinction runs alongside the first. Three different things can change when a situation changes, and a statistic can move for any of them:

- (i) the weighting measure itself may change;
- (ii) the quantity being averaged may change, with the weights untouched;
- (iii) the resulting numerical statistic may change, as a consequence of either.

For the hanging spring it is emphatically (ii) that drives most of the action. The count measure is untouched by deformation: the coils keep their identity, their number, and their mass. What changes is the function being averaged, the position $x(n)$ of each coil, and the pitch $p(n)$ between them. So the centre of mass moves although its weights never change, because the coordinate it is averaging has moved beneath it. The geometric midpoint moves because the length measure genuinely changes. An invariant measure does not guarantee an invariant average, and saying so prevents the mistake of concluding that because mass was conserved the centre of mass should have stayed put.

Quantity	Weighting	Averaged	Statistic
Coil count, total mass	count	—	unchanged
Identity of median coil	count	—	unchanged
Position of median coil	count	changes	changes
Pitch p	—	changes	changes
Geometric midpoint	length, changes	changes	changes
Centre of mass	count, unchanged	changes	changes
Weight per coil	count $\times$ field	unchanged in a uniform field	unchanged

Table 2: Separating the three ways a statistic can move under reorientation and deformation.

The last row settles whether the field can be said to have redistributed the spring's weight. In a uniform field it adds nothing, because the weight of each coil is its mass times a constant and

the weight measure is the count measure rescaled. The field changed the configuration, through the elastic response to the tension the weight produced, and thereby moved the point through which an unchanged resultant acts. Nothing got heavier. A student who says the bottom got heavier has treated weight as a possessed attribute rather than as a relation between body and field, which is the substance error in its purest form [2].

Extension: the convergent field

This section is independent of everything above and is not required by it. It treats the case in which the third measure, weight, detaches from mass.

Let the field converge as $g(r) = g_0(r_0/r)^2$, with the support at radius r_0 and the spring hanging radially inward. The weight measure $g \, dm$ is now no longer proportional to the mass measure, because g depends on where the coil is, and so a third rule for how-much has become variant. The phrase “weight distribution” now names something the mass distribution does not already contain, which in a uniform field it did not.

The dial is $\varepsilon = L/r_0$. Ignoring stretch and taking a uniform rod of length L hanging radially, the centre of mass is at $r_0 - L/2$ while the centre of gravity is the weight-weighted mean radius,

$$r_G = \frac{\int_{r_0-L}^{r_0} r \cdot r^{-2} \, dr}{\int_{r_0-L}^{r_0} r^{-2} \, dr} = r_0 \frac{(1 - \varepsilon) \ln(1/(1 - \varepsilon))}{\varepsilon}, \quad r_{\text{cm}} - r_G = \frac{L^2}{6 r_0} + O(\varepsilon^3 r_0),$$

so the centre of gravity sits closer to the attracting centre than the centre of mass. For a metre of spring on Earth this is some tens of nanometres and no laboratory will see it; for a hundred-kilometre tether hanging from an orbiting platform it is a little over 260 m, which is the scale at which tether engineers must care where the resultant acts [7]. A spring laid tangentially rather than radially is in slight compression rather than tension, because the components of the coils’ weights along its length point inward from both ends, and its pitch bunches toward its own middle rather than toward one end; the sign of that effect is a good prediction to ask for, since it needs nothing beyond resolving a weight vector.

What survives here is what survived before: the coils are still the coils, and the count measure is untouched by any field, convergent or not.

A Classroom sequence

The order below introduces the material to a class in the order a learner needs it, which inverts the order of exposition used above: the measurements come before the vocabulary, and each name arrives to label something already observed. The student investigation [3] sets it out in full with prediction prompts and recording tables.

1. Flat spring: midpoint, balance point and marked middle coil all coincide.
2. Hang it, with predictions written down before any measurement.
3. Eight blocks of equal coil count; tabulate the unequal lengths they occupy.
4. Random coil against random centimetre, making the two samplings tangible.
5. Define a measure, via μ_{count} , μ_{len} and additivity.

6. Introduce the pitch as the local conversion rate between them.
7. Derive the pitch profile from the tension.
8. Compare the centres; compare the two averages.
9. Name the conversion rate: Radon–Nikodym derivative.
10. The convergent field, separately and later.

The summary sentence at the end of the sequence is that a measure says how much weight each part of a collection receives, and an average is incomplete until the measure supplying those weights has been named.

References

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- [3] McCulloch-James, L. (2026). *How Much, Not Where: Two Ways of Measuring One Spring* [student investigation]. McMurmerings. https://leelemacjames.github.io/twoMeasures_student.pdf
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